



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Supplementary Examinations, July-2025

COMPUTER ORIENTED STATISTICAL METHODS

(COMMON TO CSE, IT, CSE (DS))

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1.
 - a State Baye's theorem. [1M] L1
 - b Define Random Variable. [1M] L1
 - c If $\text{Var}(X)=10$ then find the value of $\text{Var}(2X+8)$. [1M] L2
 - d If the mean of a Poisson variable is 1.8 then find $P(X>1)$. [1M] L2
 - e Define Uniform distribution. [1M] L1
 - f Explain Random Sampling. [1M] L1
 - g Define Standard error. [1M] L1
 - h Explain Type-I error. [1M] L1
 - i Define Markov process. [1M] L1
 - j Identify the given matrix is Stochastic or not. $\begin{bmatrix} 0 & 2 \\ 1 & 1 \\ 4 & 4 \end{bmatrix}$ [1M] L1

PART- B

(5x10=50)

- 2 a) What is the probability that a card drawn at random from the pack of playing cards may be either a queen or a king. [5M] L3
- b) State and Prove addition theorem on Probability for two events. [5M] L2

OR

- 3 A random variable 'X' has the following probability function: [10M] L3

X	0	1	2	3	4	5	6	7
P(x)	0	K	2K	2K	3K	K ²	2K ²	7K ² +K

Find (i) The value of 'K' (ii) $P(X<6)$, $P(X\geq 6)$, $P(0<X<5)$ and $P(0\leq X\leq 4)$

(iii) Determine the distribution function of X.

- 4 A continuous random variable has the probability density function $f(x)=Ax e^{-\lambda x}$, $x\geq 0$, $\lambda>0$. Determine (i) The value of 'A' (ii) Mean (iii) Variance. [10M] L3

OR

- 5 Define Binomial distribution. Out of 800 families with 5 children each, [10M] L3
how many would you expect to have (a) 3 boys (b) 5 girls (c) either 2 or 3
boys (d) at least one boy? Assume equal probabilities for boys and girls.

- 6 In a sample of 1000 cases, the mean of a certain test is 14 and standard [10M] L3
deviation is 2.5. Assuming the distribution to be normal, find (i) How
many students' score between 12 and 15? (ii) How many students score
above 18? (iii) How many students below 18?

OR

- 7 Samples of size 2 are taken from the population 1, 2, 3, 4, 5, 6. Consider [10M] L3
all possible samples drawn without replacement from this population. Find
i) Population Mean ii) Population Variance iii) Mean of Sampling
distribution of the sample mean iv) Standard error of sample mean.

- 8 Random sample of 400 men and 600 women were asked whether they [10M] L3
would like to have a flyover near their residence. 200 men and 325
women were in favour of the proposal. Test the hypothesis that
proportions of men and women in favour of the proposal are same, at 5%
level.

OR

- 9 1000 students at college level were graded according to their I.Q and the [10M] L3
economic conditions of their home. Use χ^2 test to find whether there is
any association between condition at home and I.Q.

Economic conditions	IQ	
	High	Low
Rich	460	140
Poor	240	160

- 10 An urn initially contains five black balls and five white balls. The [10M] L3
following experiment is repeated indefinitely: a ball is drawn from the
urn; if the ball is white, it is put back in the urn, otherwise it is left out.
Let X_n be the number of black balls remaining in the urn after n draws
from the urn. (a) Is X_n a Markov process? If so, find the appropriate
transition probabilities. (b) Find the one-step transition probability matrix
 P for X_n (c) Do the transition probabilities depend on n ?

OR

- 11 A gambler has Rs.2. He bets Rs.1. at a time and wins Rs.1 with [10M] L3
probability $1/2$. He stops playing if he loses Rs.2 or wins Rs. 4. (a) What
is the tpm of the related Markov Chain? (b) What is the probability that
he has lost his money at the end of 5 plays? (c) What is the probability
the game lasts more than 7 plays?

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B.Tech II Year I Semester Regular Examinations, January-2025

COMPUTER ORIENTED STATISTICAL METHODS

(COMPUTER SCIENCE AND ENGINEERING)

Max. Marks: 60

Time: 3 hours

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART-A

(10 Marks)

1. a) What is the probability for a leap year to have 52 Mondays and 53 Sundays. [1M] L2
- b) Define conditional probability with example. [1M] L1
- c) Define Poisson distribution [1M] L1
- d) If the mean and variance of a binomial distribution are 2 and 8/5. Find n. [1M] L2
- e) Find the value of the finite population correction factor for $n=10$ and $N=1000$. [1M] L2
- f) Write Central limit theorem. [1M] L1
- g) A random sample of size 81 was taken whose variance is 20.25 and mean 32. Construct 99% confidence interval. [1M] L2
- h) Define Null Hypothesis, Alternative Hypothesis. [1M] L1
- i) Find whether the matrix $\begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$ is regular or not and give reasons. [1M] L2
- j) If TPM is $\begin{bmatrix} 0 & 0.2 & x \\ x & 0.1 & y \\ 0.1 & 0.2 & z \end{bmatrix}$ find x, y, z . [1M] L2

PART-B

(5 X 10 = 50 Marks)

2. a) Out of 15 items, 4 items are not in a good condition. If 4 are selected at random find the probability that i) all are not good ii) two are not good. [5M] L3
- b) Two marbles are drawn in succession from a box containing 10 red, 30 white, 20 blue and 15 orange marbles with replacement being made after each drawing. Find the probability that i) both are white ii) first is red and second is white. [5M] L3

OR

3. a) Two dice are thrown. Let A be the event that the sum of the points on the faces is 9. Let B be the event that at least one number is 6. Find the probability of the following events [8M] L3
 - i) $P(A \cap B)$ ii) $P(A \cup B)$ iii) $P(A \cap B^c)$ iv) $P(A^c \cap B)$ v) $P(A^c \cup B^c)$ vi) $P(A/B)$ vii) $P(B/A)$ viii) $P(A^c/B)$
- b) State Baye's theorem. [2M] L2
4. a) A hospital switch board receives an average of 4 emergency calls in a 10-minute interval. [5M] L3
 - Find the probability that (a) there are at most 2 emergency calls in a 10-minute interval
 - (b) there are exactly 3 emergency calls in a 10-minute interval.
- b) Average number of accidents on any day on a national highway is 1.8. Determine the probability that the number of the accidents are i) at least one ii) at most none [5M] L3

OR

5. a) The incidence of an occupational disease in an industry is such that the workers have a 20% chance of suffering from it. What is the probability that out of 6 workers chosen at random, four or more will suffer from the disease? [5M] L3
 b) If 3 out of 20 tyres are defective and 4 of them are randomly chosen for inspection, what is the probability that only one of the defective tyre will be included? [5M] L3

6. A population consists of five numbers 2,3,4,5 and 10. Consider all possible samples of size 2 that can be drawn with replacement from this population. Find i) The mean of the population ii) The standard deviation of the population iii) The mean of the sampling distribution of means iv) The standard deviation of the sampling distribution of means. [10M] L3

OR

7. a) In a normal distribution with mean 56 and standard deviation 21, how large a sample must be taken so that there will be at least 90% chance that its mean is greater than 52. [5M] L3
 b) Prove that the mean and variance of continuous uniform distribution are $\frac{a+b}{2}$ and $\frac{(b-a)^2}{12}$ respectively. [5M] L3

8. a) Find 95% confidence limits for mean of a normality distributed population from which the following sample was taken 15, 17, 10, 18, 16, 9, 7, 11, 13, 14. [5M] L3
 b) The mean yield of wheat from a district A was 210 pounds with S.D 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 220 pounds with S.D 12 pounds from a sample of 150 plots. Assuming that the S.D of yield in the entire state was 11 pounds, test whether there is any significant difference between the mean yield of crop in the two districts. [5M] L3

OR

9. a) In a random sample of 400 industrial accidents, it was found that 231 were due at least partially to unsafe working conditions. Construct a 99% confidence interval for the corresponding true proportion. [5M] L3
 b) Random sample of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same at 5% level. [5M] L3

10. a) A person owning a scooter has the option to switch over to scooter, bike or a car next time with the probability of [0.3 0.5 0.2]. If the transition probability matrix is $\begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}$, what are the probability of vehicles related to his fourth purchase. [5M] L3
 b) Explain about gamblers Ruin Problem. Calculate the probability of ruin and expected duration of the game, when (i) $a = 100, z = 5, p = 0.6$ (ii) $a = 50, z = 40, p = 0.5$ [5M] L3

OR

11. An urn initially contains five black balls and five white balls. The following experiment is repeated indefinitely. A ball is drawn from the urn; If the ball is white it is put back in to the urn, otherwise it is left out. Let X_n be the number of black balls remaining in the urn after 'n' draws from the urn. Is X_n a Markov Process? If so a) Find the appropriate transition probabilities b) Find the one - step transition probability matrix for X_n c) Find two step transition probability matrix P^2 by matrix multiplication. [10M] L3

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PART-A

(10 Marks)

1. a) Define Mutually exclusive and independent events. [1M] L1
- b) The probability A can solve a problem is 0.8 and B can solve is 0.9, what is the probability that at least one can solve the problem. [1M] L2
- c) Define Binomial distribution. [1M] L1
- d) The probability that John hits a target is $1/3$. He fires 6 times. Find the probability that he hits the target exactly 1 time. [1M] L2
- e) When we take a sample mean from an infinite population what happens to the standard error of mean if the sample size is increased from 50 to 200. [1M] L2
- f) Write any two properties of uniform distribution. [1M] L1
- g) In a sample of 500 people in Maharashtra. 300 are wheat eaters. What can you say about the maximum error with 99% confidence. [1M] L2
- h) Define Critical Region [1M] L1
- i) Define Markov chain [1M] L1
- j) Find whether the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ is stochastic or not and give reasons? [1M] L2

PART-B

(5 X 10 = 50 Marks)

2. a) From a city 3 newspapers A, B, C are being published. A is read by 20%, B is read by 16%, C is read by 14%, both A and B are read by 8%, both A and C are read by 5%, both B and C are read by 4% and all three A, B, C are read by 2%. What is the percentage of the population that read at least one paper? [5M] L3
- b) The students in a class are selected at random, one after the other, for an examination, find the probability that the boys and girls are alternate i) if there are 5 boys and 4 girls ii) 4 boys and 4 girls. [5M] L2

OR

3. a) In a certain town 40% have brown hair, 25% have brown eyes and 15% have both brown hair and brown eyes. A person is selected at random from the town. i) If he has brown hair, what is the probability that he has brown eyes also? ii) If he has brown eyes, what is the probability that he doesn't have brown hair? [6M] L3
- b) Out of 15 bulbs, 4 bulbs are not in a good condition. If 4 bulbs are selected at random find the probability that i) all bulbs are not good ii) two bulbs are not good. [4M] L2
4. a) Suppose X is a Discrete R.V which assumes the values -3, 6, 9 with their probabilities $1/6, 1/2, 1/3$ then Find i). $E[X]$ ii). $E[2X + 3]$ iii). $V[X]$ iv). $V[2X + 3]$ [8M] L3

- b) There are 60 misprints in a book which has 300 pages. Find the probability that the page 200 has no misprints.

[2M] L2

OR

5. Fit a binomial distribution to the following data.

[10M] L3

$X=x$	0	1	2	3	4	5
$f(x)$	2	14	20	34	22	8

6. a) In a normal distribution with mean 56 and standard deviation 21, how large a sample must be taken so that there will be atleast 90% chance that its mean is greater than 52.

[5M] L3

- b) A random sample of size 100 is taken from an infinite population having the mean 76 and the variance 256. What is the probability that will be between 75 and 78.

[5M] L3

OR

7. A population consists of five numbers 2, 3, 6, 8 and 11. Consider all possible samples of size-2 that can be drawn without replacement from this population. Find: i) The mean of the population ii) The standard deviation of the population ii) The mean of the sampling distribution of means iv) The standard deviation of the sampling distribution of means (i.e., the standard error of means).

[10M] L3

8. The heights of 10 males of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. It is reasonable to believe that the average height is greater than 64 inches? Test at 5% significance level. [10M]

OR

9. The following are the number of sales which a sample of 9 sales people in city A and 6 sales people in city B made over a fixed period of time. Test whether there is any difference between the mean sales of city A and city B at 0.05 level of significance.

[10M] L3

City A	59	68	44	71	63	46	69	54	48
City B	50	36	62	52	70	41			

10. a) Three boys A, B, C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C but C is just as likely to throw the ball to B as A. Show that the process is a Markovian. Find TPM. & classify the states. Do all the states are ergodic.

[8M] L3

- b) Define a stochastic process and its state space

[2M] L2

OR

11. a) Suppose that the probability of a dry day (state 0) following a rainy day (state 1) is $1/3$ and that probability of a rainy day following a dry day is $1/2$. Given that May 1st is a dry day. Find the probability

that (i) May 3rd also a dry day. (ii) May 5th also a dry day b) Find whether the matrix $\begin{bmatrix} 0.75 & 0.25 & 0 \\ 0 & 0.5 & 0.5 \\ 0.6 & 0.4 & 0 \end{bmatrix}$ is

regular or not. Also classify states.

[8M] L3

- b) Define regular matrix and explain with one example matrix.

[2M] L3

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